SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 2 – Case Study

Course Code:	DSA03	**Course Title:**	Natural Language Processing
CO Assessed:	CO3 – Precision	**Assessment:**	Assessment Tool 2 – Case Study
Weightage:	40% of CIA		
CO5 Statement:	Apply N-gram models, smoothing techniques, part-of-speech tagging systems and to evaluate effective syntactic analysis. (BL-3)		
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Section / Batch:	B.E CSE AI / 2024	**Date:**	08-08-2026

Code of Conduct

*I **NARAPPAGARI SRAVYA(192472192)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: SRAVYA

Instructions

- Read all three case studies carefully before attempting the questions.
- Provide appropriate justifications and interpretations for every computed result.
- Use NLP concepts, algorithms, and terminology wherever applicable.
- Diagrams, flowcharts, tables, or mathematical formulations may be used to support your answers.
- Duration: 30 minutes.

Section / Topic	Focus	Case Study
N-Gram Language Models and Smoothing Techniques	Bigram and Trigram probability computation, Maximum Likelihood Estimation (MLE), Backoff models, Deleted Interpolation, Entropy calculation, uncertainty analysis, real-time next-word prediction	Case Study 1 – Smart Mobile Keyboard Prediction System
Part-of-Speech (POS) Tagging and Word Classes	Penn Treebank tagset, contextual ambiguity resolution, rule-based tagging, stochastic tagging (HMM), emission and transition probabilities, word class identification, chatbot language understanding	Case Study 2 – AI-Powered Customer Support Chatbot
Transformation-Based Tagging (Brill Tagger)	POS tag correction, transformation rules, contextual tagging, linguistic validation, tag sequence refinement, ambiguity handling	Case Study 3 – News Analytics and POS Tag Correction System
Corpus Statistics and Probabilistic Analysis	Word frequency counting, corpus-based probability estimation, entropy-based uncertainty measurement, tagging confidence evaluation	Case Study 3 – News Analytics and POS Tag Correction System
Integrated Syntactic Analysis and NLP Applications	Application of language models, POS tagging, probabilistic methods, corpus analysis, ambiguity resolution, and decision-making in real-world NLP systems	Case Studies 1, 2, and 3

Annexure A – Case Study

CASE STUDY 1: Smart Mobile Keyboard Prediction System

A smartphone company is developing an AI-powered keyboard application that predicts the next word while users type emails, messages, and social media posts. The language model is trained on the corpus:
"Data science is powerful, data science drives innovation, data science is evolving."
The development team employs Bigram and Trigram Language Models, Backoff Models, Deleted Interpolation, and Entropy Analysis to improve prediction accuracy and handle unseen word sequences. The following statistics are obtained from the corpus:

- C(data) = 3
- C(data science) = 3
- C(science is) = 2
- C(science drives) = 1
- C(science drives innovation) = 1

The interpolation weights are:

- $\lambda_1 = 0.5$ (Trigram)
- $\lambda_2 = 0.3$ (Bigram)
- $\lambda_3 = 0.2$ (Unigram)

The prediction probabilities after the word "science" are:

- P(is) = 0.66
- P(drives) = 0.33

Questions

1. Compute the Maximum Likelihood Estimate (MLE) of the Bigram Probability P(science | data). Interpret how this probability influences next-word prediction in a mobile keyboard application.
2. A user types the unseen sequence "data science improves". Apply the Backoff Model and estimate the probability using lower-order n-grams. Explain why backoff is essential for real-time predictive text systems.
3. Using Deleted Interpolation, calculate the probability of the sequence "data science is". Discuss how interpolation balances reliability and coverage in sparse datasets.
4. Calculate the Entropy of the distribution P(is)=0.66 and P(drives)=0.33. Interpret the result and evaluate how entropy helps measure uncertainty and prediction confidence in intelligent keyboard systems.

CASE STUDY 2: AI-Powered Customer Support Chatbot

A multinational airline deploys a customer support chatbot that processes user messages and automatically identifies the grammatical role of words before generating responses.

Consider the following user inputs:

1. "Book a flight ticket now."
2. "This book is interesting."

The chatbot uses the Penn Treebank POS Tagset and a Hidden Markov Model (HMM) for probabilistic tagging.

Given:

- P(book | VB) = 0.6
- P(book | NN) = 0.4
- P(Start → VB) = 0.5
- P(Start → NN) = 0.5

The system must determine whether the word "book" functions as a Verb (VB) or Noun (NN) depending on context.

Questions

1. Assign appropriate Penn Treebank POS tags to every word in both sentences. Justify the tag assigned to the word "book" using contextual clues.
2. Using the HMM probabilities, compute the likelihood of "book" being tagged as a Verb (VB) in the sentence "Book a flight ticket now." Explain why the probabilistic model favors this tag.
3. Compare Rule-Based Tagging and Stochastic (HMM) Tagging in resolving lexical ambiguity. Recommend which approach is more suitable for large-scale chatbot deployment and justify your answer.
4. Evaluate the role of standardized POS tagsets and word classes in improving intent detection, response generation, and overall chatbot performance.

CASE STUDY 3: News Analytics and POS Tag Correction System

A digital news analytics company processes millions of news articles daily. The platform uses a Transformation-Based Tagger (Brill Tagger) to improve tagging accuracy after an initial POS tagging phase.

The sentence processed by the system is:

"Economic growth increases employment."

Initial POS Tags:

- economic/JJ
- growth/NN
- increases/NNS
- employment/NN

A learned transformation rule states:

Change NNS to VBZ if the preceding word is tagged as NN.

The system also performs corpus-based frequency analysis and uncertainty evaluation.

Questions

1. Apply the transformation rule and update the POS tag sequence. Explain why the correction is linguistically appropriate.
2. Analyze the initial tagging errors and justify the corrected tag assignments using grammatical structure and sentence semantics.
3. Assuming the corpus contains the words:
    - economic (120 occurrences)
    - growth (450 occurrences)
    - increases (210 occurrences)
    - employment (380 occurrences)

    Compute the word frequency distribution and explain how frequency information supports probabilistic POS tagging.

4. Evaluate how Transformation-Based Tagging improves tagging accuracy compared with the initial tagging stage. Discuss how entropy can be used to measure uncertainty in tag assignments before and after rule application.

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

Q. No. / Criteria Level	Understanding of Case	Application of Concepts	Analysis	Solution Design	Clarity	Sub. Total	Total %
1							
2							
3							

Faculty In-charge	Course Coordinator	Program Director
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________

1. MLE of Bigram Probability P(science | data)

The MLE formula for a bigram is:

Given:

- C(data science)=3
- C(data)=3

Therefore,

Answer:

This means that in the given training corpus, **100% of the occurrences of "data" are followed by "science"**. Therefore, when a user types **"data"**, the keyboard has very high confidence in predicting **"science"** as the next word.

2. Backoff Model for "data science improves"

The sequence **"data science improves"** contains an unseen trigram:

Therefore, the trigram model cannot directly provide a probability.

The Backoff Model uses lower-order n-grams.

For the unseen word **"improves"**, the available information does not provide a count for:

or

Therefore, based strictly on the supplied corpus statistics, the exact numerical probability **cannot be computed**.

A simplified backoff interpretation is:

Since neither nor is provided, we cannot assign a verified numerical value.

Why backoff is important:

Backoff allows the keyboard to make a prediction even when a particular trigram or bigram has never appeared in the training corpus. This is essential for real-time keyboards because users constantly type **new words and previously unseen word combinations**.

3. Deleted Interpolation for "data science is"

The interpolation formula is:

Given: $P(wi|wi-2,wi-1)=\lambda 1Ptri+\lambda 2Pbi+\lambda 3Puni$

For:

data science is

Trigram probability

$$P(is|data,science)=C(data\ science) / C(data\ science\ is)$$

The corpus gives:

C(data science is)=1 and C(data)=3

and

Therefore:

Ptri=1/3=0.3333

Bigram probability

P(is|science)=C(science) / C(science is)

We know:

C(science is)=2

but **is not explicitly provided**.

Therefore, the exact bigram probability cannot be calculated from the supplied statistics.

Similarly, the unigram probability requires the total corpus word count and count of **is**, which are not fully specified.

Hence, the **exact Deleted Interpolation probability cannot be numerically determined from the given information**.

The formula would be:

$$P(is|data, science) = 0.5P_{tri} + 0.3P_{bi} + 0.2P_{uni}$$

$$= 0.5(0.3333) + 0.3P_{bi} + 0.2P_{uni}$$

$$= \boxed{0.1667 + 0.3P_{bi} + 0.2P_{uni}}$$

Interpretation: Deleted interpolation combines trigram, bigram, and unigram evidence. The trigram receives the highest weight, while lower-order models provide coverage when higher-order statistics are sparse.

4. Entropy of P(is)=0.66 and P(drives)=0.33

Entropy is:

$$H = -\sum P(x) \log_2 P(x)$$

Therefore:

$$H = -[0.66 \log_2(0.66) + 0.33 \log_2(0.33)]$$

Using the values:

$$H \approx 0.66(0.599) + 0.33(1.599)$$

$$H \approx 0.395 + 0.528$$

$$\boxed{H \approx 0.92 \; bits}$$

There is a small issue in the supplied probabilities: **0.66 + 0.33 = 0.99**, not 1.00. So 0.92 bits is the entropy using the values exactly as given.

Interpretation:

- Lower entropy → less uncertainty → more confident prediction.
- Higher entropy → greater uncertainty → less confident prediction.

Here, **"is" is considerably more likely than "drives"**, so the keyboard has a relatively strong preference for predicting **"is"** after "science".

CASE STUDY 2: AI-Powered Customer Support Chatbot

3. Penn Treebank POS Tags

Sentence 1

"Book a flight ticket now."

Word	POS Tag	Explanation
Book	VB	Verb
a	DT	Determiner
flight	NN	Noun
ticket	NN	Noun
now	RB	Adverb

Therefore:

Here **"Book" is a verb** because it expresses an action or command.

Sentence 2

"This book is interesting."

Word	POS Tag	Explanation
This	DT	Determiner
book	NN	Noun
is	VBZ	Verb
interesting	JJ	Adjective

Therefore:

Here **"book" is a noun** because it refers to an object.

Why does "book" change?

- **Book a flight** → action → **VB**
- **This book** → object/entity → **NN**

The surrounding words provide important contextual information.

4. HMM likelihood of "book" as VB

Given:

Given:

$$P(book|VB) = 0.6$$

and

$$P(Start \to VB) = 0.5$$

The simplified HMM score is:

$$P(VB, book) = P(Start \to VB) \times P(book|VB)$$
$$= 0.5 \times 0.6$$
$$= \boxed{0.30}$$

For NN:

$$P(NN, book) = 0.5 \times 0.4$$
$$= \boxed{0.20}$$

Therefore:

$$0.30 > 0.20$$

So the HMM favors:

$$\boxed{book/VB}$$

This agrees with the sentence context because **"Book a flight..."** is an imperative instruction.

5. Rule-Based vs Stochastic HMM Tagging

Feature	**Rule-Based Tagging**	**HMM Tagging**
Method	Hand-written rules	Probabilities learned from data
Ambiguity	Depends on rules	Uses contextual probabilities
Training data	Usually not required	Required
Flexibility	Limited	High
Large datasets	Difficult to maintain rules	Handles large datasets well
New contexts	May fail	Can generalize using probabilities

Recommendation

For a **large-scale chatbot**, **HMM/stochastic tagging is generally more suitable** because it can use probabilities learned from large corpora and handle ambiguous words according to context.

However, modern production chatbots would typically use **neural/transformer-based NLP models** rather than a basic HMM.

6. Importance of Standardized POS Tagsets

Standardized POS tags such as the **Penn Treebank tagset** provide a consistent representation of grammatical information.

They help the chatbot with:

1. **Intent detection** – grammatical patterns can help identify what the user wants.
2. **Response generation** – grammatical structure helps generate meaningful responses.
3. **Ambiguity resolution** – distinguishes words such as *book* used as a noun or verb.
4. **Information extraction** – nouns, verbs, adjectives, etc. can be identified.
5. **Better NLP processing** – downstream tasks receive structured linguistic information.

Thus, standardized POS tagging improves the consistency and reliability of chatbot processing.

CASE STUDY 3: News Analytics and POS Tag Correction

1. Apply the Transformation Rule

Initial tags:

economic/JJ

growth/NN

increases/NNS

employment/NN

Transformation rule:

Change NNS to VBZ if the preceding word is tagged as NN.

The word **"increases"** is initially:

Its preceding word is:

Therefore, the rule applies:

Corrected sequence

This is linguistically appropriate because **"increases" is functioning as the main verb** of the sentence.

2. Analyze the Initial Tagging Errors

Initial:

economic/JJ

growth/NN

increases/NNS

employment/NN

economic/JJ

Correct because **economic** describes the noun **growth**.

growth/NN

Correct because **growth** is the subject noun.

increases/NNS

Incorrect.

The tag **NNS** means plural noun. However, in:

Economic growth increases employment.

increases describes an action performed by the subject **economic growth**.

Therefore, it should be:

VBZ represents a **third-person singular present-tense verb**.

employment/NN

Correct because **employment** is the object of the verb **increases**.

3. Word Frequency Distribution

Given:

Word	**Frequency**
economic	120
growth	450
increases	210
employment	380

Total frequency

Therefore:

Frequency distribution

Word	**Frequency**	**Probability**
economic	120	0.1034
growth	450	0.3879
increases	210	0.1810
employment	380	0.3276

Approximately:

- economic → **10.34%**
- growth → **38.79%**
- increases → **18.10%**
- employment → **32.76%**

How frequency helps POS tagging

Corpus frequency provides statistical evidence about how commonly words occur and how they are distributed across grammatical categories.

For example, if **"increases"** frequently appears as a verb in training data, a probabilistic tagger can assign a higher probability to **VBZ** when the surrounding context supports a verb interpretation.

However, **frequency alone is not sufficient**. Context and grammatical structure are also important.

4. Transformation-Based Tagging and Entropy

Improvement in tagging accuracy

The initial tagger incorrectly assigns:

The transformation rule changes it to:

Therefore, the transformation corrects the tagging error while leaving the other correct tags unchanged.

This demonstrates how **Brill's Transformation-Based Tagging** works:

1. Perform initial tagging.
2. Identify likely errors.
3. Apply learned contextual rules.
4. Produce corrected tags.

Thus, transformation-based tagging can improve accuracy by correcting systematic contextual errors.

Entropy and uncertainty

Entropy is: $H=-i\sum P(ti)log2P(ti)$

where is the probability of a possible POS tag.

Suppose before correction the system is uncertain between:

- NNS = 0.6
- VBZ = 0.4

After applying the reliable transformation rule, suppose the system becomes highly confident in VBZ:

- VBZ = 0.9
- NNS = 0.1

Then:

Then:

$$H = -[0.9\log_2(0.9) + 0.1\log_2(0.1)]$$

$$\boxed{H \approx 0.469\ bits}$$

So:

$$0.971 > 0.469$$

So:

Interpretation: Entropy decreases after the rule is applied, indicating **reduced uncertainty and greater confidence in the POS assignment**.

Note: The entropy values above are an illustrative example because the case study does not provide actual before/after POS-tag probabilities. The exact entropy cannot be calculated without those probabilities.

Final Quick Revision

Case Study	**Main Concept**	**Key Answer**
1.1	Bigram MLE	
1.2	Backoff	Use lower-order n-grams when higher-order n-gram is unseen
1.3	Deleted Interpolation	
1.4	Entropy	≈ **0.92 bits** using supplied 0.66 and 0.33
2.3	POS Tagging	Book/VB vs book/NN based on context
2.4	HMM	VB score = **0.30**, NN score = **0.20**
2.5	Tagging methods	HMM better than basic rules for large-scale probabilistic tagging
2.6	POS Tagset	Provides standardized grammatical representation
3.1	Brill Tagger	increases/NNS → increases/VBZ
3.3	Frequency	Total = **1160**
3.4	Entropy	Lower entropy = lower uncertainty